

Van Duc Cuong

0919 968 496 | duccuongtoan@gmail.com | Hanoi, Vietnam

EDUCATION

Hanoi University of Science and Technology (HUST)

B.Sc. in Computer Science, Talented Program

- GPA: 3.82/4.00; CPA: 3.81/4.00.
- Relevant coursework: Probability and Statistics, Discrete Mathematics, Optimization Methods, and Data Structures and Algorithms (4.0).

Hanoi, Vietnam

Oct. 2022 – Present

Phan Boi Chau Specialized High School

Specialized Program in Mathematics

- Third Prize, Vietnamese Mathematical Olympiad (National Level).

Nghe An, Vietnam

2019 – 2022

RESEARCH EXPERIENCE

VinMotion

AI Researcher

- Research Vision-Language-Action models, foundation models, and scalable data generation for embodied AI and humanoid-robot learning.
- Develop foundation-model-based approaches for whole-body locomotion and manipulation, long-horizon control, perception, and embodied intelligence.
- Investigate robot-learning frameworks that combine multimodal reasoning, synthetic data, and large pretrained models for complex tasks.

Hanoi, Vietnam

2025 – Present

BKAI Laboratory, Hanoi University of Science and Technology

Undergraduate Research Assistant

- Develop and analyze evolutionary algorithms and machine-learning methods for combinatorial optimization and automated algorithm design.
- Investigate hybrid frameworks that integrate large language models with evolutionary optimization for heuristic discovery, complex search, and decision-making.
- Study reflective, feedback-driven mechanisms that iteratively improve algorithmic solutions generated by LLMs.

Hanoi, Vietnam

2023 – Present

FPT Software AI Center

AI Resident Researcher

- Collaborated with Tien's Lab at The University of Texas at Dallas on LLMs for software engineering, including regression testing, program reasoning, and specification inference.
- Developed execution-aware, multi-turn LLM frameworks for automated test generation and program analysis.

Hanoi, Vietnam

2025 – 2026

PUBLICATIONS

Conference Papers

Van Duc Cuong, Nguyen Dinh Tuan Minh, Vu Duc Tam, Vu Duy Tung, Nguyen Van Son, Nguyen Thi Hanh, and Huynh Thi Thanh Binh. "ReVEL: Multi-Turn Reflective LLM-Guided Heuristic Evolution via Structured Performance Feedback." *Findings of EMNLP, 2026. CORE A**.

Nguyen Le Minh Hoang*, **Van Duc Cuong***, Bui Trong Duc, and Huynh Thi Thanh Binh. "COMPASS: Composable Policy-Amortized Structured Search for LLM-Based Optimization Modeling." *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), 2026. CORE A**.

Cuong Chi Le*, **Van Duc Cuong***, Tung Duy Vu, Minh V. T. Pham, Hoang N. Phan, Huy N. Phan, and Tien N. Nguyen. "TestWeaver: Execution-aware, Feedback-driven Regression Testing Generation with Large Language Models." *ICSE, 2026. CORE A**.

Cuong Chi Le, Minh V. T. Pham, Tung Duy Vu, **Van Duc Cuong**, Huy N. Phan, Hoang N. Phan, and Tien N. Nguyen. "SpecMind: Cognitively Inspired, Interactive Multi-Turn Framework for Postcondition Inference." *ACL, 2026. CORE A**.

Ha Minh Hieu, Hung Phan, Duc Chinh Tran, **Van Duc Cuong**, Dao Van Tung, and Huynh Thi Thanh Binh. "Alimentation Deep Multiple Optimal Ant Colony Optimization for the Vehicle Routing Problem with Time Windows." *GECCO Companion, 2024. CORE A*.

Van Duc Cuong, Tran Quoc Thai, Nguyen Dinh Tuan Minh, Vu Duc Tam, Nguyen Van Son, and Nguyen Thi Hanh. "MiRAGE: Misconception Detection with Retrieval-Guided Multi-Stage Reasoning and Ensemble Fusion." *SOICT, 2026. CORE B*.

Van Duc Cuong, Ta Dinh Tam, Tran Duc Chinh, and Nguyen Thi Hanh. "FLUID: Flow-Latent Unified Integration via Token Distillation for Expert Specialization in Multimodal Learning." *SOICT, 2026. CORE B*.

Journal Articles

Huynh Thi Thanh Binh, **Van Duc Cuong**, Nguyen Van Son, La Van Quan, and Nguyen Thi Hanh. “LSHADE-NGS: Enhancing Q-Coverage in Directional Sensor Networks through Navigated Generation Search.” *Neural Computing and Applications*, 2025. **Q1 Journal**.

La Van Quan, Tran Quoc Thai, **Van Duc Cuong**, Nguyen Dinh Tuan Minh, Nguyen Van Son, Nguyen Thi Hanh, and Huynh Thi Thanh Binh. “ALSB: A Problem-Aware Hybrid Optimization Framework for k-Coverage and k-Connectivity in Wireless Sensor Networks.” *Knowledge-Based Systems*, 2026. **Q1 Journal**.

Nguyen Thanh Cong, Nguyen Van Son, Nguyen Ngoc Hao, **Van Duc Cuong**, Nguyen Minh Hieu, Nguyen Tien Dung, and Nguyen Thi Hanh. “AutoMind: Automated Insight Discovery via Multi-Agent Navigation.” *IEEE Access*, 2026. **Q1 Journal**.

Nguyen Thi Hanh, **Van Duc Cuong**, Tran Duc Chinh, Ha Minh Hieu, Nguyen Van Son, and La Van Quan. “Memetic Algorithm for Maximizing K-Coverage and K-Connectivity in Wireless Sensor Networks.” *Informatica*, 2024. **ISI Q2**.

* Equal contribution.

HONORS & AWARDS

Best Presentation Award , Hanoi University of Science and Technology	2026
Academic Achievement Scholarship (Type A) , HUST	2024 – 2025
Third Prize , University-Level Scientific Research, HUST	2024
Third Prize , ChatGPT-Solver Competition (AI for SAT Solving), HUST	2023
Third Prize , Vietnamese Mathematical Olympiad (National Level)	2022
Second Prize , Informatics Excellence Competition, Nghe An Province	2022

RESEARCH INTERESTS

Large Language Models; Vision-Language-Action Models; Embodied AI; Humanoid Robotics; Evolutionary Computation; Automated Algorithm Design; Combinatorial Optimization; AI for Software Engineering.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, LaTeX

Machine Learning: PyTorch, Deep Learning, Large Language Models, Vision-Language Models, Reinforcement Learning

Robotics: ROS/ROS2, MuJoCo, Isaac Sim, Robot Learning, Simulation, Whole-Body Control

Research: Evolutionary Computation, Combinatorial Optimization, Automated Algorithm Design, Scientific Writing

REFERENCE

Prof. Huynh Thi Thanh Binh

Professor and Vice Dean, School of Information and Communication Technology (SoICT)

Hanoi University of Science and Technology (HUST)

users.soict.hust.edu.vn/binhht/index.htm